Phase diagram of networks grown with triadic closure rules

Alexei Vazquez[1, *]

[1] *Nodes & Links Ltd, Salisbury House, Station Road, Cambridge, CB1 2LA, UK*

(Dated: October 22, 2026)

We map numerically, up to $n = 10^7$ nodes, the phase diagram of a network grown by four independent processes: nodes arrive at rate α_+ and attach to a random node, random nodes are deleted at rate $\alpha_- < \alpha_+$, open triads close at rate β_+, and closed triads open at rate β_-. For $\beta_- \geq \beta_+$ the triangles form a critical or subcritical queue, and the network is a tree with a Poisson-like degree tail. For $\beta_+ > \beta_-$ the outcome depends on how the open triad is found. We take as order parameter the fraction $P_\triangle$ of triangles in the largest cluster of triangles that share links. When triads are drawn uniformly from the whole network, the closure links concentrate on a vanishing fraction of the nodes. Without opening they form a near-clique of about $\sqrt{n}$ nodes that holds nearly all the triangles, and this condensate survives node deletion as a steady state rebuilt from young nodes. Opening thins it. At one net closure link per arrival and beyond a turnover β_-/β_+ between 0.76 and 0.78, the triangles stop percolating up to $n = 10^7$ nodes, although the closure links still concentrate on the nodes of highest degree; whether this dispersed regime is a phase or a long crossover remains open. Deletion moves the boundary towards $\beta_- = \beta_+$. Closure within a node's neighbourhood never condenses.

I. INTRODUCTION

Triadic closure, a link forming between two nodes that share a neighbour, is the mechanism behind the high clustering of social networks [1–6]. In an evolving university e-mail network, pairs two steps apart were about 30 times more likely to form a tie than pairs three steps apart, and the probability of a new tie grew with the number of common contacts [7]. In dynamic online networks such as Digg and Facebook wall posts the growth is consistent with a linear dependence on that number [8], which is what follows when every open triad closes at the same rate, as under rule (w) below. This evidence concerns the rate at which a pair closes; it does not test the rate at which a node introduces its friends, which under rule (w) grows as the square of its degree and feeds the hubs of Sec. VI. In the local-search model $LS(n, \ell)$ [4, 9, 10] with $\ell = 1$ a newcomer picks a random node j, links to it and to a random neighbour of j. Every arrival closes one triangle, so arrival and closure are the same event, and no link is ever lost.

In a social system these are separate processes. People join and leave at their own rates, acquaintances are introduced to each other at another, and ties lapse at yet another. We therefore let four processes run side by side: arrival at rate α_+, deletion of a node at rate α_-, closure of an open triad at rate β_+, and opening of a closed triad, the reverse of closure, at rate β_-. Meetings biased towards pairs with mutual friends, combined with the decay of unrenewed ties, give clustered networks with strong communities [11], and growth by triadic closure alone generates communities, which heterogeneous fitness replaces with a few superhubs [12]. On a fixed population, link creation through friends of friends balanced by random decay gives a discontinuous transition between a sparse and a dense society [13], and a birth rate that grows with the number of common neighbours gives bistability [14]. Closure through a uniformly random wedge, our rule (w), defines the wedge picking model of Ahmadian and Haddadan [8]. They bounded the growth of a vertex's degree over short time windows on a fixed vertex set; we follow the same step over long times in a growing network. Static triadic closure on a fixed backbone is exactly solvable [15], and in the random triadic process the graph becomes complete above a sharp threshold [16].

Growth changes the outcome. In a fixed population the dense and sparse states compete globally; in a growing network nodes differ by age, the dense state appears on a vanishing fraction of them, and bistability becomes condensation.

Our results are the following. A stationary density requires total rates in fixed ratio, and it fixes the mean degree; at $\beta_- = \beta_+$ the triangles become a critical queue and the network a tree (Sec. III). When closure draws its triads from the whole network, the triangles condense, after a plateau whose length grows quickly as the closure rate decreases, onto a near-clique of about $\sqrt{n}$ nodes that survives node deletion (Sec. VI). Opening thins the condensate into a sparse cluster and, beyond a turnover of about 0.77, stops the triangles from percolating up to $n = 10^7$, although closure links still gather on the hubs (Secs. V and VII); we cannot exclude a later onset. Closure rules that find the triad locally never condense (Sec. IX). The mean field (Sec. IV) misses all of this, and communities are discussed in Sec. X.

II. THE MODEL

The network starts from a path of three nodes. With $Z = \alpha_+ + \alpha_- + \beta_+ + \beta_-$, each event is an *arrival* with

* alexei@nodeslinks.com

probability α_+/Z, in which a new node links to a uniformly random node; a *deletion* with probability α_-/Z, in which a uniformly random node is removed with all its links; a *closure* with probability β_+/Z, in which an open triad a–c–b (a, b unlinked) gets the link (a,b); or an *opening* with probability β_-/Z, in which a closed triad loses the link (a,b). Since Z only sets the clock, rates are measured in units of α_+. We keep $\alpha_- < \alpha_+$ so that the network grows; for $\alpha_- > \alpha_+$ its size stays small.

How a triad is chosen defines the rule, and each closure rule has a mirror-image opening rule. Under the *random-triad* rule (w) an open (closed) triad is drawn uniformly from the whole network. Every open triad thus closes at the same rate, so an unlinked pair with m common neighbours links at a rate proportional to m, and a node with k neighbours introduces each of its $k(k-1)/2$ pairs at the same rate. The centre of an open triad has weight $k_c(k_c-1)$, and opening removes a link with probability proportional to its number of triangles, t_e. Under the *random-potential-link* rule (p), closure picks a uniformly random unlinked pair with a common neighbour, whatever their number of common neighbours, and opening a uniformly random link with $t_e \geq 1$; we sample an open triad as in rule (w) and accept it with probability $1/m$, where m is the number of common neighbours of its pair, so that every eligible pair is equally likely; in the dense core most draws are rejected, which costs time but introduces no bias. Under the *introducer* rule (c), a random node with $k_c \geq 2$ links (unlinks) two of its neighbours. Under the *friend-of-a-friend* rule (f), a random node a walks $a \to j \to l$, $l \neq a$, and links to (unlinks from) l. Rules (w) and (p) need a census of the network; (c) and (f) use only a node and its neighbourhood. A selection that does not qualify is repeated. A closure that finds no open triad becomes an arrival, and an opening that finds no closed triad does nothing; for $\beta_- < \beta_+$ this happens only in the first few events, and for $\alpha_- = 0.95\,\alpha_+$ in 4×10^{-5} of the closures. Opening never disconnects the network, since a and b stay joined through c; deletion can. Rule (w) is the main object of this paper; rules (p), (c) and (f) are compared with it in Sec. IX, with $LS(n,1)$, one node and two links per step, as reference.

The simulation keeps adjacency lists, a hash set of links, and Fenwick trees over the triad weights and, with opening, over the t_e. Triangle counts are updated incrementally and were checked against full recounts. Each link is tagged with the event that created it. A run stops when the network first holds n nodes, so runs with the same seed and different n are stages of one trajectory.

III. DENSITY AND THE FIRST TRANSITION

Write $r_\pm = \beta_\pm/\alpha_+$ for the number of closures and openings per arrival, and

$$\Delta = r_+ - r_- = \frac{\beta_+ - \beta_-}{\alpha_+} \tag{1}$$

for the net closure per arrival, the links that closure adds, net of opening, per arrival. A deletion removes a node and, on average, $\langle k\rangle$ links. If every opening finds a closed triad, the links grow at rate $\alpha_+ + \beta_+ - \beta_- - \alpha_-\langle k\rangle$ and the nodes at rate $\alpha_+ - \alpha_-$, and a stationary density $E/n = \langle k\rangle/2$ requires

$$\langle k\rangle = \frac{2(1+\Delta)}{1+\alpha_-/\alpha_+}. \tag{2}$$

The rates must therefore be held in fixed ratio. Without opening or deletion, $\beta_+ = \alpha_+$ gives $\langle k\rangle = 4$, the density of $LS(n,1)$. All runs agree with Eq. (2) to within 2%.

Per-triad rates, with which every open triad closes at its own constant rate, define a different, densifying model, like those grown by copying and triadic linking [17, 18]; for rule (w) its mean field has a stationary density only up to $\beta_+/\alpha_+ = 0.067$ (Sec. S1 of Ref. [19]). We study total rates, at fixed density.

In the rest of this section $\alpha_- = 0$.

Equation (2) cannot hold for $\beta_- \geq \beta_+$, since it would give $\langle k\rangle \leq 2$ while the network stays connected, $E \geq n-1$. The openings must then fail often enough to keep $E \geq n-1$. Close to a tree each closure creates one triangle and each opening destroys one. The triangle count is therefore a queue with arrival rate β_+ and service rate β_-. It grows linearly in n for $\beta_+ > \beta_-$, is critical at $\beta_+ = \beta_-$, and stays $O(1)$ for $\beta_- > \beta_+$. In the last case a fraction $(\beta_- - \beta_+)/\beta_-$ of the openings find nothing to remove.

The simulations follow this. At $\beta_+ = \alpha_+$ and $n = 10^6$ the fraction of failed openings matches $(\beta_- - \beta_+)/\beta_-$ to within 0.002 at $\beta_-/\beta_+ = 1.1$, 2 and 10, and at $\beta_+ = \beta_-$ the triangles grow as $\sqrt{n}$ (Sec. VIII). The plane $\beta_- = \beta_+$ thus separates extensive triangles from a tree. Deletion removes triangles but never creates them, so this boundary holds at any α_-, the tree becoming a forest.

IV. UNCORRELATED MEAN FIELD

Here $\alpha_- = 0$. Let a closure pick q of its two endpoints with probability proportional to degree and the other $2-q$ uniformly. In the absence of correlations, the degree of a node grows with the number of arrivals as

$$n\frac{dk_i}{dn} = 1 + r_+(2-q) + \frac{r_+ q}{\langle k\rangle}k_i - r_-\frac{2t_i}{3T/n}, \tag{3}$$

where the last term is opening under rule (w): node i is an endpoint of the removed link with probability $2t_i/3T$, with t_i its number of triangles. Rules (w), (c) and (p) reach both endpoints through a link, so $q = 2$. Rule (f) starts from a uniformly random node, so $q = 1$.

Without opening, with $u = \beta_+/(\alpha_+ + \beta_+)$, so that $r_+ = u/(1-u)$ and $\langle k\rangle = 2/(1-u)$, Eq. (3) gives $k_i \sim (n/n_i)^b$ with $b = q\,u/2$, and

$$\gamma = 1 + \frac{1}{b} = 1 + \frac{2}{q\,u}. \tag{4}$$

At $u = 1/2$, rules (w), (c) and (p) give $\gamma = 3$, and rule (f) gives $\gamma = 5$. This is also the exponent of $LS(n, 1)$, in which each arrival adds one link to a uniformly random node j and one to a random neighbour of j, reached with probability $k_i/(n\langle k\rangle)$ in the uncorrelated limit: $n\,dk_i/dn = 1 + k_i/4$ at $\langle k\rangle = 4$.

With opening under rule (w), if $t_i \propto k_i$, Eq. (3) becomes $n\,dk_i/dn = 1 + (1 - 2/\langle k\rangle)k_i$, independent of β_- at fixed density. In the simulations ($\Delta = 1$, $n = 10^6$) $t_i \propto k_i^{1+\sigma}$ with $\sigma = 0.84$, 0.39, 0.06, -0.37 at $\beta_-/\beta_+ =$ 0.5, 0.67, 0.8, 0.89: the correlations carry the effect of turnover.

V. THE PHASE DIAGRAM

Condensation is a statement about where the triangles go, and the order parameter should say so without a choice of scale. Call two triangles adjacent when they share a link, and a *triangle cluster* a connected set of adjacent triangles: a 3-clique community in the language of clique percolation [20, 21]. Every closure creates at least one triangle, so the clusters record where the closures went. Let T be the number of triangles in the network and $T_{\max}$ the number of triangles in its largest triangle cluster. We take as order parameter the fraction of all triangles in the largest cluster,

$$P_\triangle = \frac{T_{\max}}{T}, \tag{5}$$

the analogue for triangles of the fraction P of nodes in the giant component, together with the fractions $P_\bullet$ and $P_{\bullet-\bullet}$ of all nodes and links that this cluster spans. If the triangles are scattered over finite clusters, $P_\triangle \to 0$. If $P_\triangle$ stays finite they percolate, and $P_\bullet$ separates extended percolation, $P_\bullet$ finite, from condensation, $P_\bullet \to 0$. The clusters are found by union-find over links, joining the three links of every triangle.

We ran rule (w) at $\Delta = 1/2$ and 1, at $\alpha_-/\alpha_+ = 0$ and 1/2, for β_-/β_+ from 0 to 0.95 and $n = 10^4$, 10^5, 10^6, refined the turnover near the boundary in steps of 0.02, and scanned $\alpha_-/\alpha_+ = 0.1$–0.9 at $\Delta = 1$ (Fig. 1). At small turnover $P_\triangle$ rises to one with n; at large turnover it falls below 10^{-3} at 10^6. We call a point condensed when $P_\triangle$ rises between 10^5 and 10^6 (or exceeds 0.9) and dispersed when it falls. Along every line of fixed Δ and α_-/α_+ the label changes once, which brackets the boundary (Table S1 of the Supplemental Material [19]). At $\Delta = 1$ the brackets are 0.76–0.78, 0.88–0.90 and 0.90–0.95 at $\alpha_-/\alpha_+ = 0$, 0.5 and 0.8; at $\alpha_-/\alpha_+ = 0.9$ every turnover up to 0.95 is condensed. At $\Delta = 1/2$ they are 0.72–0.74 and 0.88–0.90 at $\alpha_-/\alpha_+ = 0$ and 0.5. Away from $\alpha_- = 0$ the brackets rest on this criterion alone, with one or two seeds at 10^6, and carry the caveat discussed next.

A trend between 10^5 and 10^6 cannot tell a dispersed point from one that condenses later, since without opening condensation follows a long plateau (Sec. VI). We therefore followed $\beta_-/\beta_+ = 0.70$–0.85 at $\Delta = 1$, $\alpha_- = 0$ along single trajectories to $n = 10^7$, and to 3×10^7 at 0.78 (Fig. S1 and Table S3 of Ref. [19]). At 0.76, $P_\triangle$ is flat between 10^5 and 10^6 (0.140 ± 0.014 and 0.137 ± 0.019 over 32 and 16 seeds) and then rises to 0.30 at 10^7: it condenses late. At 0.80 and 0.85, $P_\triangle \le 10^{-4}$ at 10^7 and T/n still decreases by 8–9% per decade. At 0.78 the precursors follow those at 0.76 about a decade later: T/n levels off and rises slightly between 10^7 and 3×10^7, and the share of links among the 1% highest-degree nodes reaches 0.31 and 0.42, against 0.36 at 0.76 and 10^7. Yet $P_\triangle$ keeps falling, to 2×10^{-4} at 3×10^7, while at 0.76 it stays above 0.03 at every size. The boundary thus lies between 0.76 and 0.78 up to these sizes, and the precursors at 0.78 suggest that it may drift to larger turnover at larger n. Near it, the scaled variance $n\,\mathrm{var}(P_\triangle)$ over 16 to 64 seeds peaks at 0.76 and grows with n, and a Binder-type cumulant dips from 0.66 to below 0.45 (Table S2 of Ref. [19]). Three sizes support no scaling collapse, and the data neither show a discontinuity nor establish a continuous transition.

Deletion moves the boundary towards $\beta_- = \beta_+$, with the uncertainty of these brackets. At $\Delta = 1$ a fit to the midpoints of the nine brackets at $\alpha_-/\alpha_+ = 0$–0.8 gives

$$1 - \left(\frac{\beta_-}{\beta_+}\right)_c \simeq 0.223\left(1 - \frac{\alpha_-}{\alpha_+}\right), \tag{6}$$

which misses two brackets by at most 0.006. It vanishes at $\alpha_- = \alpha_+$ by construction, and the brackets there are too wide to tell whether the dispersed regime closes. The shift is not an effect of density or of the rates per surviving node: at $\alpha_-/\alpha_+ = 0.5$ the boundary is at 0.89 for both $\Delta = 1/2$ and 1, whereas at the same $\langle k\rangle \approx 3$ it is at 0.73 ($\Delta = 1/2$, $\alpha_- = 0$) and 0.875 ($\Delta = 1$, $\alpha_-/\alpha_+ = 0.3$). We have not identified its mechanism.

There are thus three regimes. In the *condensed* phase one triangle cluster, made of closure links, holds essentially all the triangles and spans a vanishing fraction of the nodes (Sec. VI). In the *dispersed* regime the triangles are extensive but stay in finite clusters up to $n = 10^7$ (Sec. VII). In the *tree* phase, $\beta_- \ge \beta_+$, there are $O(1)$ triangles (Sec. VIII). Their degree distributions are in Fig. 2b.

VI. THE CONDENSED PHASE

A. Without opening: a dense core

With $\beta_- = \alpha_- = 0$ the only parameter is $u = \beta_+/(\alpha_+ + \beta_+)$, and $\langle k\rangle = 2/(1 - u)$. Under rule (w) at $u = 1/2$ and $n = 10^6$, the $\sqrt{n} = 10^3$ highest-degree nodes are linked to one another with density 0.81 and hold 20% of all links, a share that does not shrink with n. Measures that do not presuppose the core size agree on $\Theta(\sqrt{n})$ nodes: between $n = 10^4$ and 10^6 the maximal k-core grows as $n^{0.47}$, the densest subgraph found by greedy peeling [23] as $n^{0.50}$, and the largest set of top-ranked nodes linked with density at least 1/2 as $n^{0.53}$. Hence

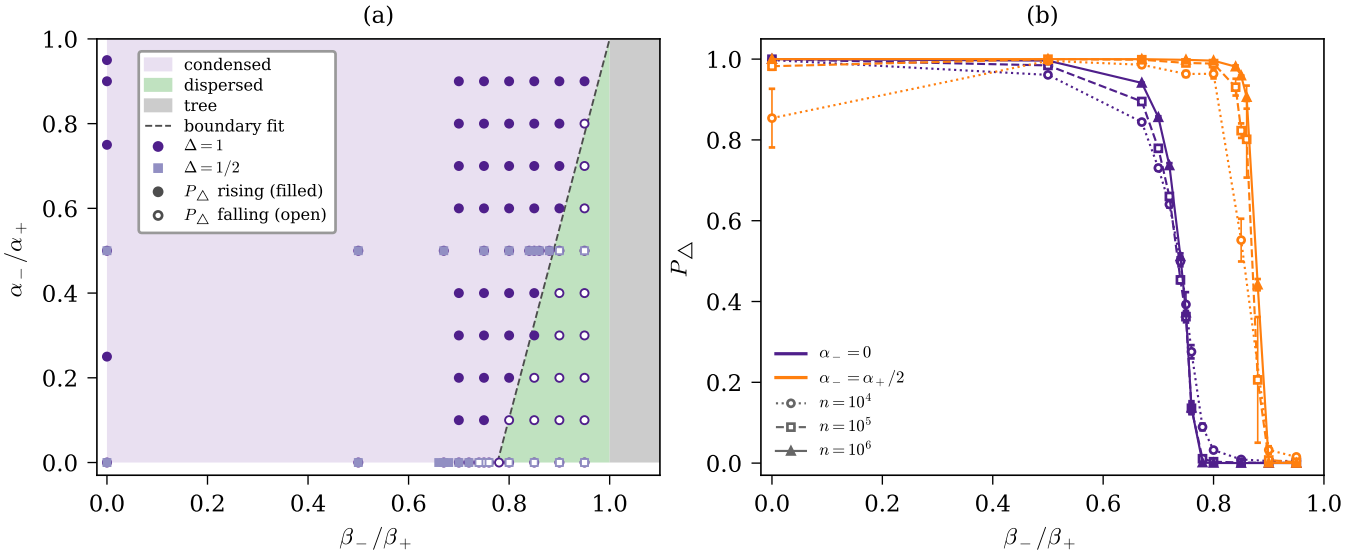


FIG. 1. Phase diagram of rule (w) from the triangle-cluster order parameter $P_\triangle$. (a) The plane of turnover β_-/β_+ and deletion α_-/α_+, at net closure per arrival $\Delta = 1$ (circles) and $\Delta = 1/2$ (squares). A symbol is filled when $P_\triangle$ rises between $n = 10^5$ and 10^6, or exceeds 0.9 at 10^6, and open when it falls; seeds per point in Table S1 of the Supplemental Material [19]. Shades: condensed (purple), dispersed (green) and tree (grey, $\beta_- \geq \beta_+$) regimes. Dashed line: the fit of Eq. (6) to the $\Delta = 1$ boundary. (b) $P_\triangle$ against β_-/β_+ at $\Delta = 1$, without deletion (purple) and with $\alpha_- = \alpha_+/2$ (orange), for $n = 10^4$ (dotted lines, open circles), 10^5 (dashed lines, open squares) and 10^6 (solid lines, filled triangles). Means over 1–6 seeds, and over 64, 32 and 16 more at $\beta_-/\beta_+ = 0.70$–0.80, $\alpha_- = 0$; error bars, one standard error.

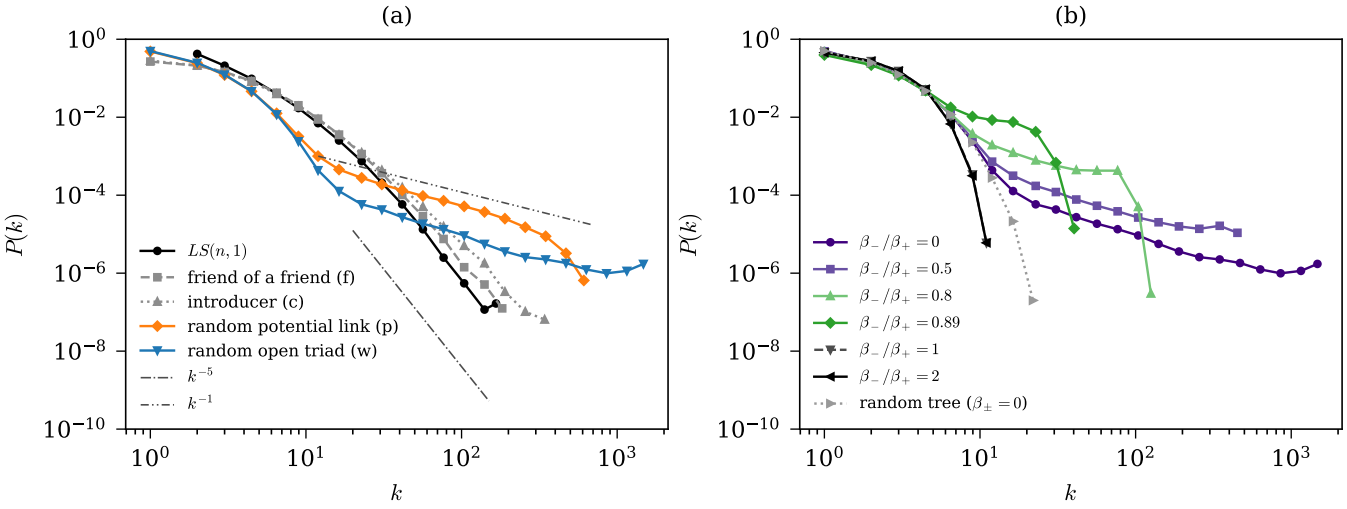


FIG. 2. Degree distributions in logarithmic bins, $n = 10^6$. (a) The closure rules without opening at $\langle k \rangle = 4$ ($u = 1/2$). Reference lines: k^{-5}, the mean-field tail of rule (f) and $LS(n, 1)$, and k^{-1}, a guide to the slow decay of rule (w) before its maximum at $k \sim \sqrt{n}$. (b) Rule (w) across the phase diagram at $\langle k \rangle = 4$: $\beta_-/\beta_+ = 0$ and 0.5 (condensed, purple), 0.8 and 0.89 (dispersed, light and dark green); at the tree transition $\beta_-/\beta_+ = 1$ (dashed) and in the tree phase, $\beta_-/\beta_+ = 2$ (black), both at $\beta_+ = \alpha_+$; and the uniform random tree, $\beta_\pm = 0$ (dotted). One network per curve.

$T \sim n^{3/2}$. The core seeds the giant triangle cluster, which holds $P_\triangle = 0.99996$ of the triangles on $P_\bullet \sim n^{-0.45}$ of the nodes (Fig. 4a): the core and the peripheral nodes tied to it by closures. Tagging each link with its origin, the cluster contains 99.4% of the closure links and 1% of the arrival links. The transitivity stays near 0.5 while the mean local clustering vanishes (Table I), and the degree distribution has a second maximum at $k \sim \sqrt{n}$ (Fig. 2a).

The hub degree follows from the core's own triads. Once triads centred in the core dominate the census, a core node a is an endpoint of a closure with probability w_a/W. Here $w_a \simeq \rho\sqrt{n}\,K$ counts the open triads

through its $\rho\sqrt{n}$ core neighbours, and $W \simeq \sqrt{n}\,K^2/2$. Hence $n\,dK/dn \simeq 2\rho r_+ n/K$ and

$$K^2 \simeq c\,r_+\,n, \tag{7}$$

with c of order 4ρ. The data give $k_{\max}^2/(r_+ n) = 2.5$–2.8 for $u = 0.3$–0.6 at $n = 10^6$; between 10^5 and 10^7, $k_{\max} \sim n^{0.48}$, close to but slightly below $\sqrt{n}$.

Onset. At small u a long plateau, with constant T/n, precedes the condensate and ends abruptly (Fig. 3a). The plateau level itself falls with u, from 0.46 at $u = 0.25$ to 0.05 at $u = 0.05$, so we take the onset size n^* as the first n at which T/n reaches twice its plateau minimum. Then $\ln n^* = 14.2$, 13.4, 11.2, 9.3 at $u = 0.08$, 0.10, 0.15, 0.25, with seed-to-seed spread 0.2–0.4 (Fig. 3b). Over $0.08 \le u \le 0.3$ a power law, $n^* \propto u^{-4.0}$, and an essential singularity, $\ln n^* = 7.1 + 0.59/u$, fit equally well (rms 0.25 and 0.22 in $\ln n^*$), but they part at smaller u: at $u = 0.05$ they predict $n^* \approx 8 \times 10^6$ and 1.6×10^8. Of two trajectories followed to 5×10^7, one condenses at $n^* \approx 5 \times 10^7$ ($\ln n^* = 17.7$) and the other has not yet, although its T/n is rising. The onset thus grows faster than the power law and, from one trajectory, no faster than the essential singularity. Condensation occurs down to $u = 0.05$ at least; the data do not decide whether it occurs at every $u > 0$ or below a threshold $u_c < 0.05$.

Mechanism. A hub of degree K is the centre of $\sim K^2/2$ triads, and each closure centred on it raises the degree of two of its neighbours. A two-variable caricature captures this feedback, which the mean field of Eq. (3) misses. Let $s = \sum_{c\in\partial a} k_c$ be the summed degree of the neighbours of a hub a of degree K, $x = \ln n$, and $\epsilon = r_+/\bar{w}$, with $\bar{w} = W/n$ the open triads per node. The hub is an endpoint of a closure with probability $w_a/W \simeq s/W$, so $K' = n\,dK/dn \simeq \epsilon s$. It is the centre of about $K^2/2$ open triads, each closure of which adds one to the degree of two of its neighbours, so $s' \gtrsim \epsilon K^2$. Hence $K'' \simeq \epsilon^2 K^2$, which blows up at a finite $x^* \sim \epsilon^{-1} K_0^{-1/2}$ for any $\epsilon > 0$. The blow-up stops when the core dominates W, which gives Eq. (7). The caricature explains why there is no threshold in sight, but not the form of $n^*(u)$. On the plateau, $\ln n^*$ against $1/\epsilon$ fits worse (rms 0.56) than either form above.

TABLE I. Rules without opening at $u = 1/2$ ($\langle k\rangle = 4$), $n = 10^6$. b: age exponent, fitted over $10^2 < n_i < 10^5$; $b_{\rm MF} = qu/2$; $\mathcal{T}$: transitivity; $\bar{C}$: mean local clustering; r_k: degree assortativity [22]; $\phi(\sqrt{n})$: club fraction of the $\sqrt{n}$ highest-degree nodes; ρ: link density among them. [a]With a core of old hubs and a second maximum at $k \sim \sqrt{n}$, $k_i(n_i)$ is not a power law for rule (w).

rule	b	$b_{\rm MF}$	$k_{\max}$	$\mathcal{T}$	$\bar{C}$	r_k	$\phi(\sqrt{n})$ (ρ)
$LS(n,1)$	0.27	1/4	169	0.25	0.69	0.05	0.001 (0.002)
(f)	0.31	1/4	203	0.24	0.37	0.28	0.002 (0.008)
(c)	0.37	1/2	391	0.22	0.42	0.16	0.002 (0.007)
(p)	0.66	1/2	684	0.03	0.02	0.52	0.026 (0.11)
(w)	—[a]	1/2	1619	0.48	0.008	0.65	0.20 (0.81)

B. With deletion: a steady state

In a growing network the hubs are the oldest nodes, so the condensate might be a relic of the first arrivals. Deletion separates the two, since every node is eventually removed. The network holds $n \propto (\alpha_+ - \alpha_-)t$ nodes, each deleted at rate α_-/n, so a node that arrived when a fraction s of all arrivals had taken place survives with probability $s^{\alpha_-/(\alpha_+-\alpha_-)}$, and the median s of the survivors is $2^{-(1-\alpha_-/\alpha_+)}$.

We ran rule (w) without opening at $u = 0.3$ and $1/2$ and $\alpha_-/\alpha_+ = 0$–0.95, up to $n = 10^6$. The condensate survives at every deletion rate. At $u = 1/2$ and $n = 10^6$, $P_\triangle \ge 0.9998$, the cluster holds 95–99% of the closure links and at most 1% of the arrival links (from the link tags), and $P_\bullet \sim n^{-0.45}$ as without deletion (Fig. 4a). The largest degree still grows as $\sqrt{n}$: $k_{\max} = 71$, 268, 986 at $n = 10^4$, 10^5, 10^6 for $\alpha_- = 0.95\,\alpha_+$. At $u = 0.3$ deletion delays the onset but does not prevent it. The condensate is young: the median arrival fraction s of the $\sqrt{n}$ highest-degree nodes is 0.001 without deletion and 0.34, 0.90, 0.96 at $\alpha_-/\alpha_+ = 0.5$, 0.9, 0.95, against 0.71, 0.93, 0.97 for a random surviving node. At strong deletion the hubs are as young as the population. The condensate is a steady state: core nodes are deleted, and census closure promotes others from the periphery to replace them.

C. With opening: a sparse cluster

Opening hits a link in proportion to its triangles, so it falls mainly on the core, which thins out: at $\Delta = 1$ and $n = 10^6$ the share of links among the $\sqrt{n}$ richest nodes falls from 0.20 to 0.02 and 0.005 at $\beta_-/\beta_+ = 0.5$ and 0.67. The triangle cluster does not dissolve. At $\beta_-/\beta_+ = 0.5$, $P_\triangle = 0.96$, 0.98, 0.996 at $n = 10^4$, 10^5, 10^6, and the cluster holds 97% of the closure links (Fig. 4b). It spreads over more nodes, but their share still vanishes, $P_\bullet \sim n^{-\lambda}$ with $\lambda = 0.44$, 0.35, 0.25 at $\beta_-/\beta_+ = 0$, 0.5, 0.67 (Fig. 4a), and the second maximum of the degree distribution disappears (Fig. 2b). Approaching the boundary the cluster first loses closure links and then triangles: at $\beta_-/\beta_+ = 0.72$ it holds 38% of them while $P_\triangle = 0.73$ (Fig. 4b).

VII. THE DISPERSED REGIME

Above the boundary the triangles break into finite clusters, but the closure links still go to the nodes of highest degree. At $\Delta = 1$, $\alpha_- = 0$ and $\beta_-/\beta_+ = 0.8$ there are 0.14 triangles per node at $n = 10^6$, while the largest cluster spans about 50 nodes (Fig. 4a, green). The share of links among the 1% highest-degree nodes grows along the trajectories as in the condensed phase (Sec. V), and

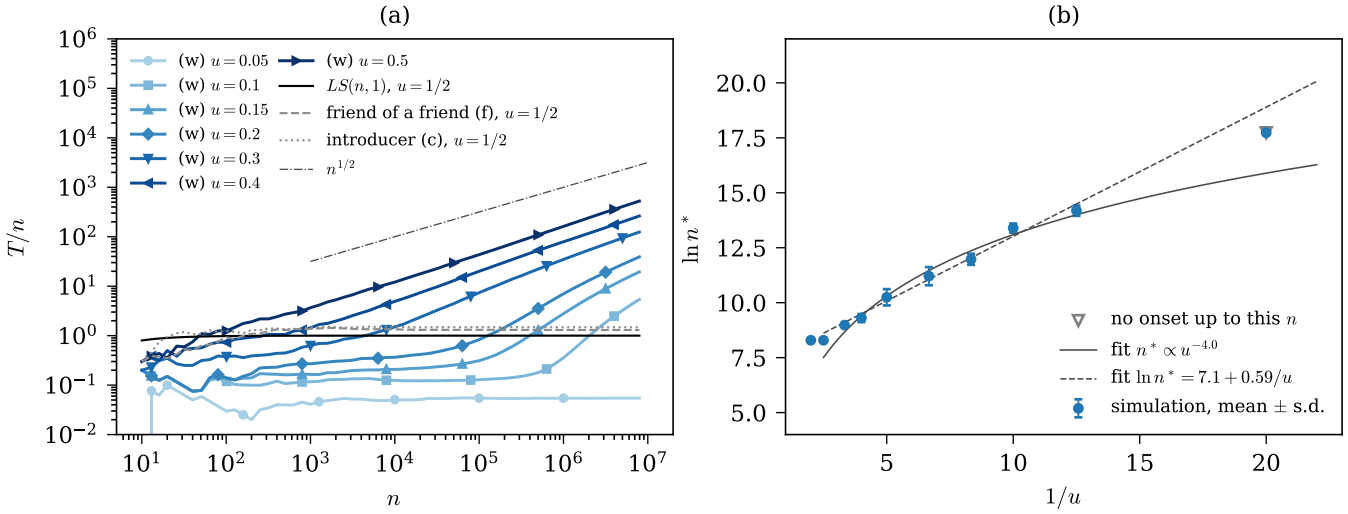


FIG. 3. Condensation without opening. (a) Triangles per node T/n along single trajectories to $n = 10^7$: rule (w) at $u = 0.05$–0.5 (blue lines with symbols, light to dark), and the local rules and $LS(n, 1)$ at $u = 1/2$ (grey and black). Dash-dotted line, $n^{1/2}$. (b) Onset size n^*, the first n at which T/n reaches twice its plateau minimum, mean and standard deviation over 5–6 seeds for $0.08 \le u \le 0.25$, two at $u = 0.05$ and one otherwise. Lines are fits over $0.08 \le u \le 0.3$: a power law (solid) and an essential singularity (dashed). At $u = 0.05$, one trajectory condenses at $n \approx 5 \times 10^7$ (circle) and the other has no onset up to that size (triangle).

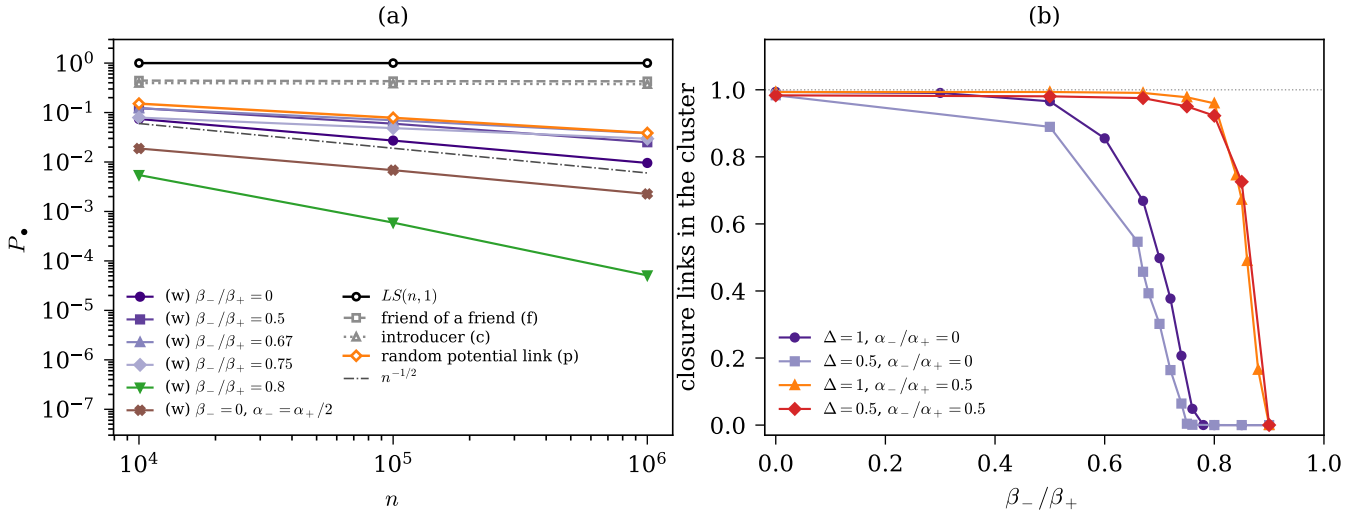


FIG. 4. The largest triangle cluster. (a) Fraction $P_\bullet$ of the nodes that it spans, against n: rule (w) at $\Delta = 1$ without deletion for $\beta_-/\beta_+ = 0$–0.75 (condensed, purple, dark to light) and 0.8 (dispersed, green); rule (w) at $\beta_- = 0$, $\Delta = 1$, $\alpha_- = \alpha_+/2$ (brown crosses); and, without opening at $u = 1/2$, rule (p) (orange diamonds), the local rules (grey) and $LS(n, 1)$ (black). Dash-dotted line, $n^{-1/2}$. Means over two seeds at $n = 10^4$ and one or two at larger n. (b) Measured fraction of the closure links that lie inside the cluster, from the link-origin tags, against β_-/β_+ at $n = 10^6$ (one seed), for $\Delta = 1$ and 1/2 with $\alpha_-/\alpha_+ = 0$ and 0.5. Dotted line: all closure links.

the largest degree grows as $n^{0.25}$ at 0.80. These hubs form the shoulder of $P(k)$ at $k \approx 30$–100 (Fig. 2b). At $\beta_-/\beta_+ = 0.8$ and $n = 10^6$ the nodes with $k \ge 30$ are 2.8% of the total, and 97% of the links among them come from closures, but their mutual link density is only 0.002 and their triangles do not connect. The same club forms in the condensed phase, where its triangles percolate: the regimes differ in whether the triangles among the hubs connect, not in whether closure links gather on them.

VIII. THE TREE PHASE

For $\beta_- \ge \beta_+$ nearly every triangle is opened soon after it forms, and two times in three the opening removes a link at the centre c, which passes it to one of its neigh-

bours. Under rule (w) the centre is drawn with weight $k_c(k_c-1)$. A node therefore loses links at a rate quadratic in its degree and gains them, as a neighbour of a centre, at a rate linear in it. The degrees are not bounded, but their tail falls faster than exponentially.

The data confirm this (Fig. 2b; Table S4 of Ref. [19]). At $\beta_+ = \alpha_+$ the largest degree grows from 9 at $n = 10^4$ to 12–14 at 10^7, against 24 for the uniform random tree at 10^6, and $k\,P(k)/P(k-1) \approx 2$ for $k \geq 4$. This Poisson-like tail, $P(k) \propto 2^k/k!$, is the one expected from losses quadratic and gains linear in k; its maximum grows as $\ln n/\ln\ln n$, by a factor 1.4 over these sizes, as measured. At $\beta_+ = \beta_-$ the degree distribution is the same, and $T = 74$ and 2060 at $n = 10^4$ and 10^7, $T \propto n^{0.50}$, as for a critical queue. Deletion adds a service term, $3T/n$ triangles per deleted node, which vanishes for $T \ll n$: it leaves the boundary in place and confines T at the boundary to the same $\sqrt{n}$ scale.

IX. OTHER CLOSURE RULES

Rule (p) also condenses, although without a dense core. At $u = 1/2$ its largest triangle cluster holds $P_\triangle = 0.97$, 0.98, 0.99 of the triangles at $n = 10^4$, 10^5, 10^6 and, at 10^6, 98% of the closure links (from the link tags) on a share of the nodes that falls as $P_\bullet \sim n^{-0.29}$ (Fig. 4a). Its core is sparse, with density 0.1 among the $\sqrt{n}$ richest nodes, and its largest degree grows as about $n^{0.35}$: without the m-fold weight of pairs sharing m neighbours there is no drive towards a clique. Its age exponent, $b = 0.66$, is far from the mean-field 1/2.

The local rules never condense. Their triangles percolate on an extended cluster: at $u = 1/2$ the largest one holds $P_\triangle = 0.83$ (f) and 0.77 (c) of the triangles and spans $P_\bullet = 0.43$ and 0.37 of the nodes at $n = 10^6$, nearly as at 10^4. In $LS(n,1)$ one cluster holds everything, $P_\triangle = P_\bullet = 1$. Without opening the local rules are described by Eq. (4) only roughly: the age exponent b is 0.31 for (f) (mean field 1/4) and 0.37 for (c) (mean field 1/2), against 0.27 for $LS(n,1)$ (Table I). With $k_{\max} \approx 200$–400 at $n = 10^6$ their tails span little more than a decade, which is too short to call them scale-free; they are broad and fall off much faster than those of the census rules (Fig. 2a). With opening at $\langle k\rangle = 4$, rule (f) keeps $b = 0.31$–0.34 up to $\beta_-/\beta_+ = 0.8$ and stays in the $LS(n,1)$ class, while rule (c) develops a broader tail without concentrating links. At $\beta_+ = \beta_-$ rule (f) reaches the same critical tree as rule (w).

X. COMMUNITIES

Triadic closure generates communities [11, 12], and in data it is hard to tell apart from community structure and homophily [26, 27]. We fit with graph-tool [28] the planted-partition model [24], which describes assortative groups only, and the nested degree-corrected stochastic block model (SBM) [25, 29], which describes any mixing. Both select the number of groups by minimum description length; on configuration-model rewirings of our networks they find one group (once the SBM finds two, with negligible gain). The fits are limited to $n = 10^5$ by memory.

The local rules and $LS(n,1)$ split into about 10^3 assortative groups, with $Q = 0.90$–0.95, a number of groups nearly proportional to n and 7 to 9 SBM levels, a hierarchy as in Refs. [11, 12] and the footprint of local closure rather than of homophily [27]. For rules (w) and (p) the SBM finds two to six groups with at most three levels; one group holds 66–95% of the nodes and the others the rich nodes, of mean degree 8 to 39. This is the club and periphery that condensation implies, and that the closure-link club of Sec. VII extends into the dispersed phase; it disappears at $\beta_-/\beta_+ = 0.94$. The planted-partition count is not stable: between seeds it jumps from 2 (club and periphery) to 20–40 small groups at the same parameters, so its non-monotonic dependence on β_-/β_+ is noise between two fits of nearly equal description length.

XI. CONCLUSIONS

Once the clock is fixed the model has three parameters: Δ, β_-/β_+ and α_-/α_+. At $\beta_- = \beta_+$ the triangles form a critical queue, and above it the network is a tree. Below it, under rule (w), the closure links gather on a vanishing fraction of the nodes, a near-clique of about $\sqrt{n}$ nodes without opening, which survives deletion. Local closure rules never condense. Unlike Bose–Einstein

TABLE II. Communities at $n = 10^5$ (in parentheses, $n = 10^4$). $B_{\rm pp}$: groups of the planted-partition model [24]; Q: modularity of that partition; S_1: fraction of nodes in its largest group; $\bar{k}_s$: mean degree outside the largest group; $B_{\rm sbm}$ and L: groups at the bottom level and nontrivial levels of the nested degree-corrected SBM [25]. Rule (w) at $\langle k\rangle = 4$, three seeds: ranges of $B_{\rm pp}$, $B_{\rm sbm}$ and L, means of Q, S_1 and $\bar{k}_s$.

network	$B_{\rm pp}$	Q	S_1	$\bar{k}_s$	$B_{\rm sbm}$	L
$LS(n,1)$	1870 (216)	0.95	0.002	4.0	2435	9
(f), $u = 1/2$	1265 (154)	0.90	0.005	4.0	1863	7
(c), $u = 1/2$	1049 (148)	0.92	0.005	4.0	2128	9
(p), $u = 1/2$	3 (8)	0.43	0.94	36	3	1
(w), $\beta_-/\beta_+ = 0$	22–41	0.27	0.95	39	2–5	1–2
(w), $\beta_-/\beta_+ = 0.5$	2–35	0.35	0.90	25	3	1
(w), $\beta_-/\beta_+ = 0.67$	2–9	0.39	0.92	28	3–6	1–3
(w), $\beta_-/\beta_+ = 0.8$	2–3	0.43	0.79	12	2–4	1–2
(w), $\beta_-/\beta_+ = 0.89$	2	0.37	0.66	8	2–3	1–2
(w), $\beta_-/\beta_+ = 0.94$	1 (1)	0	1	—	1	0
(w), $\beta_+ = \beta_-$	1 (1)	0	1	—	1	0
random tree	1 (1)	0	1	—	1	0

condensation [30] or superlinear preferential attachment [31], where one node captures a finite fraction of the links, here a growing set of nodes does.

The main open question is whether the dispersed regime survives as $n \to \infty$. Without opening the onset of condensation recedes quickly as the closure rate decreases, and above the boundary the closure links still gather on the hubs up to $n = 10^7$. A correlated theory of this club, with opening as a loss that grows with its triangles, would decide it and give the onset $n^*(u)$ and the order of the transition.

ACKNOWLEDGMENTS

All code is available at Ref. [32]. The calculations were assisted by Claude Opus 5, an AI assistant developed by Anthropic. Nodes & Links Ltd provided support in the form of salary for Alexei Vazquez but did not have any additional role in the conceptualization of the study, the analysis, the decision to publish or the preparation of the manuscript.

[1] M. S. Granovetter, The strength of weak ties, Am. J. Sociol. **78**, 1360 (1973).
[2] J. Davidsen, H. Ebel, and S. Bornholdt, Emergence of a small world from local interactions: Modeling acquaintance networks, Phys. Rev. Lett. **88**, 128701 (2002).
[3] P. Holme and B. J. Kim, Growing scale-free networks with tunable clustering, Phys. Rev. E **65**, 026107 (2002).
[4] A. Vazquez, Growing network with local rules: Preferential attachment, clustering hierarchy, and degree correlations, Phys. Rev. E **67**, 056104 (2003).
[5] R. Toivonen, J.-P. Onnela, J. Saramäki, J. Hyvönen, and K. Kaski, A model for social networks, Physica A **371**, 851 (2006).
[6] M. O. Jackson and B. W. Rogers, Meeting strangers and friends of friends: How random are social networks?, Am. Econ. Rev. **97**, 890 (2007).
[7] G. Kossinets and D. J. Watts, Empirical analysis of an evolving social network, Science **311**, 88 (2006).
[8] S. Ahmadian and S. Haddadan, The wedge picking model: A theoretical analysis of graph evolution caused by triadic closure and algorithmic implications, J. Strateg. Innov. Sustain. **16**(3), 74 (2021); arXiv:2012.01691.
[9] A. Vazquez, Disordered networks generated by recursive searches, Europhys. Lett. **54**, 430 (2001).
[10] A. Vazquez, *Local network growth: how simple rules drive network complexity* (Independently published, 2026), ISBN 979-8181138933.
[11] E. M. Jin, M. Girvan, and M. E. J. Newman, Structure of growing social networks, Phys. Rev. E **64**, 046132 (2001).
[12] G. Bianconi, R. K. Darst, J. Iacovacci, and S. Fortunato, Triadic closure as a basic generating mechanism of communities in complex networks, Phys. Rev. E **90**, 042806 (2014).
[13] M. Marsili, F. Vega-Redondo, and F. Slanina, The rise and fall of a networked society: A formal model, Proc. Natl. Acad. Sci. USA **101**, 1439 (2004).
[14] P. Grindrod, D. J. Higham, and M. C. Parsons, Bistability through triadic closure, Internet Math. **8**, 402 (2012).
[15] L. Cirigliano, C. Castellano, G. J. Baxter, and G. Timár, Strongly clustered random graphs via triadic closure: An exactly solvable model, Phys. Rev. E **109**, 024306 (2024).
[16] D. Korándi, Y. Peled, and B. Sudakov, A random triadic process, SIAM J. Discrete Math. **30**, 1 (2016).
[17] U. Bhat, P. L. Krapivsky, R. Lambiotte, and S. Redner, Densification and structural transitions in networks that grow by node copying, Phys. Rev. E **94**, 062302 (2016).
[18] R. Lambiotte, P. L. Krapivsky, U. Bhat, and S. Redner, Structural transitions in densifying networks, Phys. Rev. Lett. **117**, 218301 (2016).
[19] See Supplemental Material at [URL] for per-triad rates (Sec. S1), the boundary brackets (Table S1), seed statistics (Table S2), trajectories to $n = 10^7$ (Fig. S1, Table S3) and the tree phase (Table S4).
[20] I. Derényi, G. Palla, and T. Vicsek, Clique percolation in random networks, Phys. Rev. Lett. **94**, 160202 (2005).
[21] G. Palla, I. Derényi, I. Farkas, and T. Vicsek, Uncovering the overlapping community structure of complex networks in nature and society, Nature **435**, 814 (2005).
[22] M. E. J. Newman, Assortative mixing in networks, Phys. Rev. Lett. **89**, 208701 (2002).
[23] M. Charikar, Greedy approximation algorithms for finding dense components in a graph, in *Approximation Algorithms for Combinatorial Optimization*, LNCS Vol. 1913 (Springer, Berlin, 2000), p. 84.
[24] L. Zhang and T. P. Peixoto, Statistical inference of assortative community structures, Phys. Rev. Research **2**, 043271 (2020).
[25] T. P. Peixoto, Hierarchical block structures and high-resolution model selection in large networks, Phys. Rev. X **4**, 011047 (2014).
[26] A. Asikainen, G. Iñiguez, J. Ureña-Carrión, K. Kaski, and M. Kivelä, Cumulative effects of triadic closure and homophily in social networks, Sci. Adv. **6**, eaax7310 (2020).
[27] T. P. Peixoto, Disentangling homophily, community structure, and triadic closure in networks, Phys. Rev. X **12**, 011004 (2022).
[28] T. P. Peixoto, The graph-tool python library, figshare, doi:10.6084/m9.figshare.1164194 (2014).
[29] B. Karrer and M. E. J. Newman, Stochastic blockmodels and community structure in networks, Phys. Rev. E **83**, 016107 (2011).
[30] G. Bianconi and A.-L. Barabási, Bose-Einstein condensation in complex networks, Phys. Rev. Lett. **86**, 5632 (2001).
[31] P. L. Krapivsky, S. Redner, and F. Leyvraz, Connectivity of growing random networks, Phys. Rev. Lett. **85**, 4629 (2000).
[32] https://github.com/av2atgh/LocalNetworkGrowth, directory `triadic_closure`.